

NE 511: Multi-Physics of Nuclear Reactors

UNIT 2: Short-Time Multi-Physics Phenomena in Reactor Core

Lecture 11: *Delayed Reactivity Feedback Phenomena:*
Moderator / Coolant Feedback Effects

MODERATOR TEMPERATURE EFFECT

The feedback effects associated with the moderator temperature may be larger in magnitude than the fuel temperature feedback effects.

Moderator temperature effect; Void effect; Pressure effect

The moderator coefficient of reactivity is given by $\alpha_m(T_m) = \frac{\Delta k}{\Delta T_m}$.

Changes in the moderator temperature ΔT_m are related to changes in the neutron flux $\Delta \phi$ by a suitable set of heat-transfer equations for the fuel lattice.

The simplest type is: $\tau_m \frac{\partial}{\partial t} \Delta T_m = \frac{\Delta \phi}{\phi^} T_m^* - \Delta T_m$*

T_m^ and ϕ^* are the initial values of moderator temperature and flux.*

In general, a weak coupling would appear at low values of T_m^ .*

MODERATOR TEMPERATURE EFFECT

The change in reactivity per degree change in moderator temperature is called moderator temperature coefficient of reactivity:

- *The magnitude and sign (+ or -) of the moderator temperature coefficient is primarily a function of the moderator-to-fuel ratio.*
- *If a reactor is under-moderated, it will have a negative moderator temperature coefficient → Increasing the moderator temperature will decrease the moderator-to-fuel ratio.*
- *If a reactor is over-moderated, it will have a positive moderator temperature coefficient → Decreasing the moderator temperature will increase the moderator-to-fuel ratio.*

Calculation of the moderator temperature reactivity coefficient for thermal reactors:

$$k = k_{\infty}P = \eta_T f p \varepsilon P$$

$\eta_T \text{ and } \varepsilon \text{ are not functions of } T_{mod}$

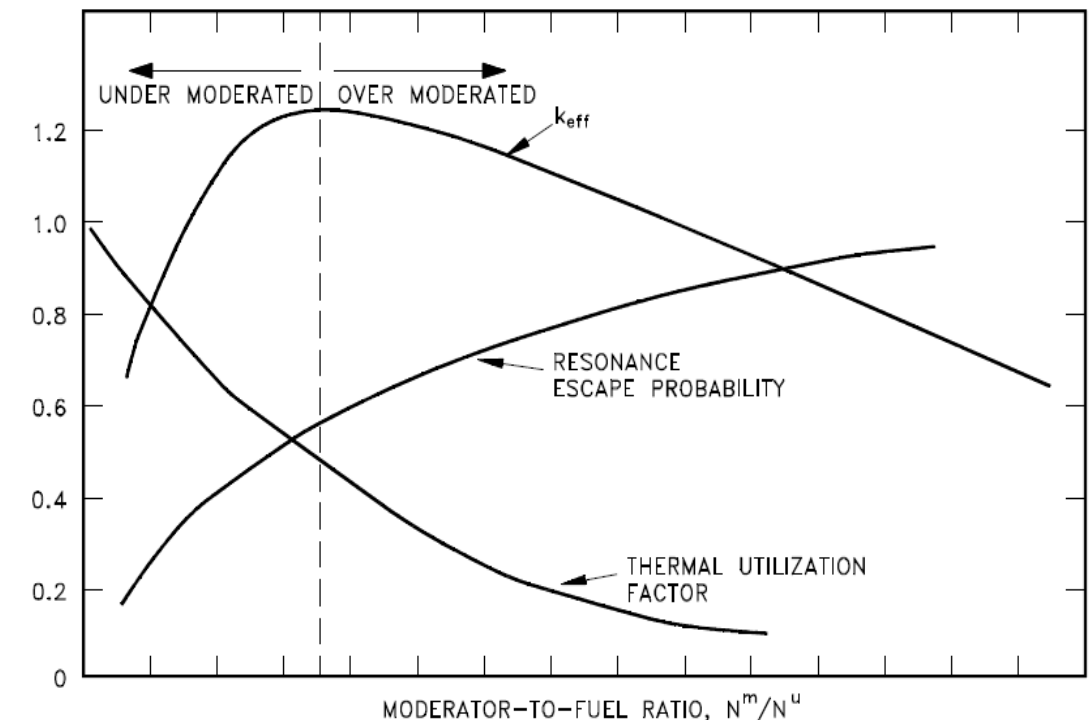
taking logarithms & differentiating
with respect to T

$a_{mod} = \alpha_T(f) + \alpha_T(p) + \alpha_T(P)$

MODERATOR TEMPERATURE EFFECT

The *moderator-to-fuel ratio* is very important in the discussion of moderators.

- As the reactor designer increases the amount of moderator in the core, neutron leakage decreases.
- If the neutron absorption in the moderator, Σ_{am} , increases it will cause a decrease in the thermal utilization factor.
- Having insufficient moderator in the core causes an increase in slowing down time and results in a greater loss of neutrons by resonance absorption → This also causes an increase in neutron leakage.
- A reactor is under-moderated when a decrease in the moderator-to-fuel ratio decreases k_{eff} due to the increased resonance absorption.
- A reactor is over-moderated when an increase in the moderator-to-fuel ratio decreases k_{eff} due to the decrease in the thermal utilization factor.



MODERATOR TEMPERATURE EFFECT

In thermal reactors, temperature changes in the moderator introduce changes in neutron multiplication through a variety of mechanisms.

Water-Moderated Reactors

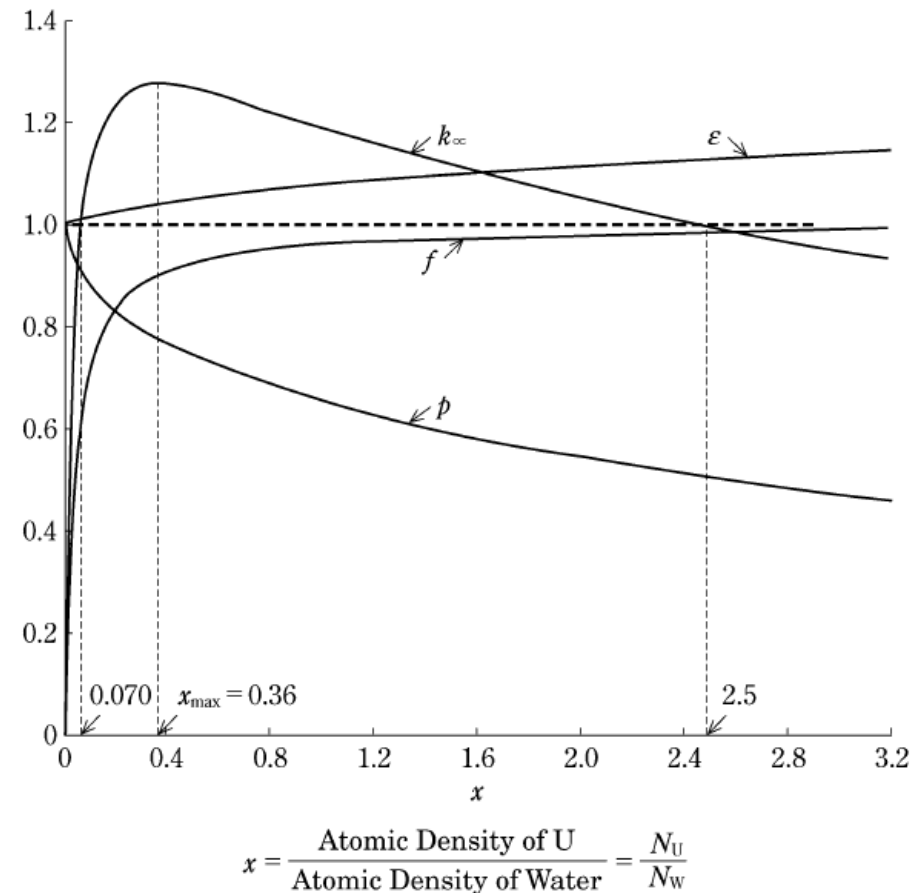
- !! In PWR, an expansion of the water leads to an expulsion of water from the core and thus to a loss of moderator.
- !! Since a PWR will generally be under-moderated, the loss of moderator will lead to a fall in the multiplication factor.
- !! Typically for a PWR the moderator temperature coefficient of reactivity is about (-10^{-4}) per °C.

MODERATOR TEMPERATURE EFFECT

Moderator expansion effect on p and f :

It can be shown that the moderator temperature coefficient *is negative if $f > p$*

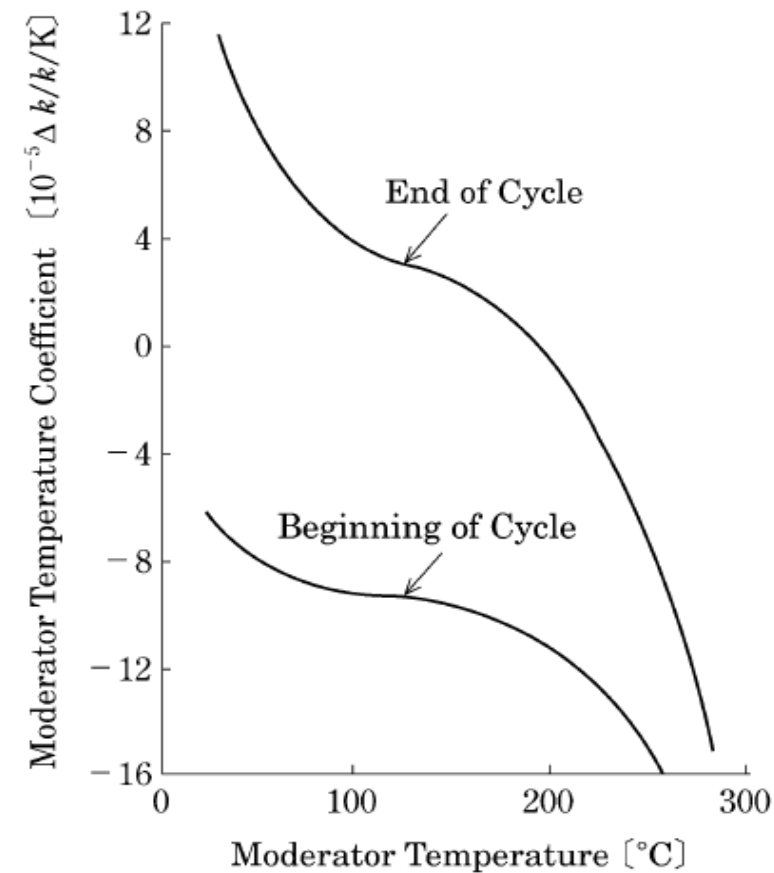
Fuel-to-moderator ratio effect on k_{∞} :



The variation of k_{∞} due to an increase in moderator temperature is determined by p and f (η is out of the range of this figure, but will not change with ϵ if there are no resonance effects)

MODERATOR TEMPERATURE EFFECT

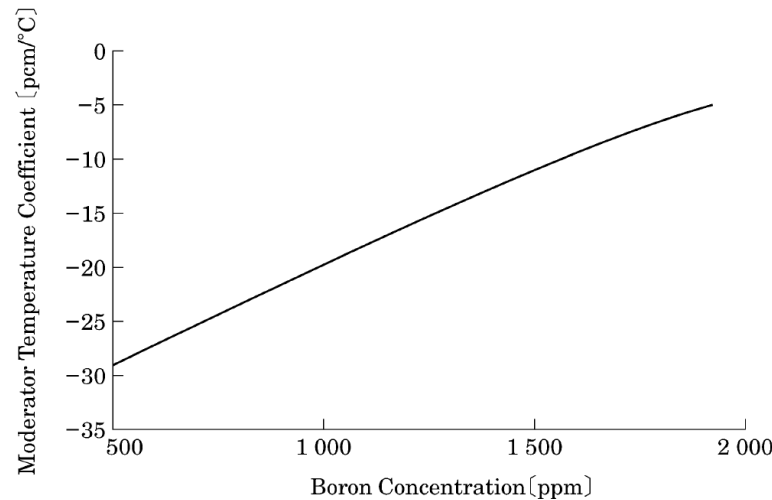
Example: shift of moderator temperature coefficient to positive with fuel burnup



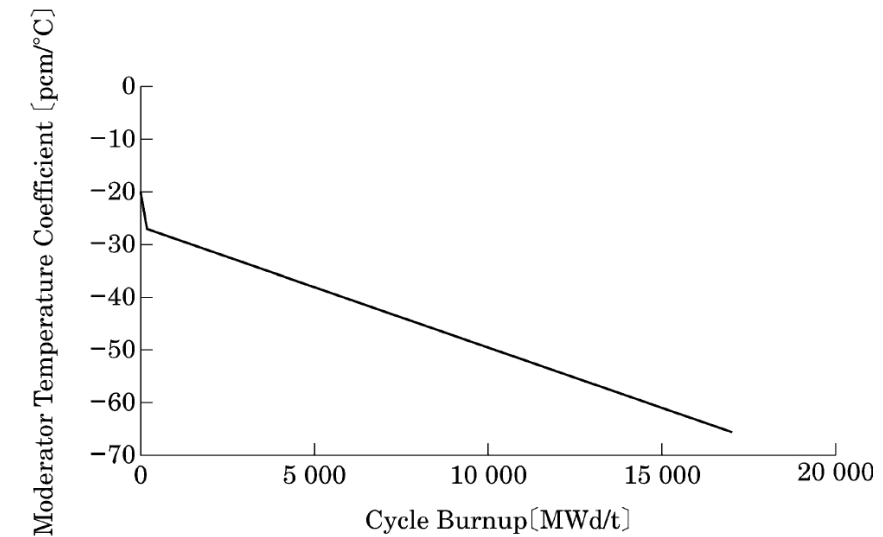
The thermal absorption rate of fissile nuclides should be large to make the reactivity effect negative against ^{239}Pu buildup

MODERATOR TEMPERATURE EFFECT

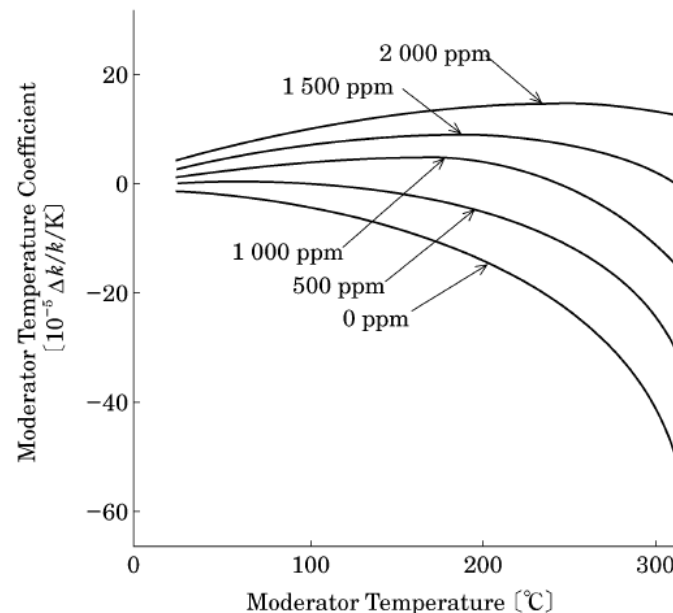
- Moderator temperature feedback and corresponding coefficient in a PWR depends also on the amount of soluble boron (shim) in the moderator (coolant).



Typical dependence of moderator temperature coefficient on boron concentration



Typical dependence of moderator temperature coefficient on cycle burnup



- If a chemical shim is present, a decrease in coolant density due to temperature rise will also lead to a decrease in boric acid concentration
- Thus, it has a positive reactivity effect on moderator temperature coefficient
- The requirement for a negative moderator temperature coefficient will limit the amount of boric acid concentration allowed at any time during the cycle depletion

MODERATOR TEMPERATURE EFFECT

Graphite-Moderated Reactors

Thermal expansion of the graphite has an insignificant effect on the moderator temperature coefficient of reactivity (small and negative effect):

- !! *With graphite-moderated reactors the most important result of changing moderator temperature is that the kinetic energy of the atoms is changed → the energy spectrum of neutrons in the reactor is changed.*
- !! *Changes in temperature will result in changes in the number of fission reactions taking place relative to other neutron reactions, such as capture and leakage.*

MODERATOR TEMPERATURE EFFECT

Graphite-Moderated Reactors

Effect of ^{239}Pu content

- !! *The cross-section of the fissile isotope ^{239}Pu has a large resonance at 0.3 eV, which is between 5 and 10 times the mean neutron energy.*
- !! *An increase in moderator temperature leads to an increase in mean neutron energy and thus to an increase in the number of absorptions (and hence fissions) in ^{239}Pu .*
- !! *Consequently, the moderator temperature coefficient becomes positive and increases in magnitude with the addition of ^{239}Pu to the fuel.*
- !! *This trend does not increase indefinitely, since with high concentration of Pu there is already a large fraction of the neutrons absorbed in ^{239}Pu and increases in resonance absorption must be at the expense of more efficient absorption: **The fission/absorption ratio is higher at lower energies.***

MODERATOR TEMPERATURE EFFECT

Graphite-Moderated Reactors

With natural uranium fueled Magnox reactors, the moderator temperature coefficient of reactivity of unirradiated fuel is roughly $\alpha_m = -2 \times 10^{-5}$ per $^{\circ}\text{C}$.

Magnox reactors operate with an on-load refueling schemes. After 2- or 3-years of operation (fuel is discharged at about 3000 MWD/t), the moderator temperature coefficients of reactivity becomes $\alpha_m = +10 \times 10^{-5}$ per $^{\circ}\text{C}$.

- ✱ Positive moderator temperature coefficient greater than the negative fuel temperature coefficient - serious difficulties in the reactor control?

Thankfully, the fuel temperature changes following power changes are considerably larger and much more rapid than the moderator temperature changes.

But nevertheless, the existence of a significant positive feedback effect is of great importance in the graphite-moderated reactor and has in many instances determined the type of control system required for this reactor.

MODERATOR TEMPERATURE EFFECT

Relevant calculations require accurate techniques for computing the neutron energy spectrum:

- 1. The neutron energy spectrum dependence on position in the lattice cell must be considered.*
- 2. Accurate knowledge of nuclear data involving both the neutron-absorbing characteristics of the fuel and the neutron-scattering properties of the moderator are needed.*

These calculations are normally compared with measurements in clean, heated lattices before application to the power reactor situation.

Experimental values for the coefficient may also be derived from experience on specially designed experiments in power reactors (accuracy may not be good).

MODERATOR TEMPERATURE EFFECT

For [graphite lattices](#) the moderator time constant, τ_m , is much larger than the fuel time constant, τ_f , due to the large thermal capacity of the moderator bricks.

The coupling between moderator temperature and neutron flux in such reactors is fairly strong.

In [pressure tube reactors \(CANDU\)](#), the moderator is thermally insulated from the reactor coolant and is maintained near constant temperature (under 100°C) by itself being circulated through a heat exchanger.

Consequently, the coupling between neutron flux and main moderator temperature is weak.

Coupling is also *weak* in [gas-cooled reactor](#), where sleeves around the fuel element thermally insulate the moderator from the coolant flowing over the fuel.

VOIDAGE EFFECTS

The voidage effects are important in BWRs.

The effect is similar to the effects of coolant temperature changes in a PWR in that both cause changes in moderator density:

The change in the number of water molecules affects slowing down fission, capture, and leakage properties of the reactor lattice.

The net effect is a complex interaction between them all, which may have either sign.

VOIDAGE EFFECTS

The response time of voidage to changes in power is a matter of only a few seconds.

It is desirable to have a small negative void coefficient of reactivity so that the control system does not have to stabilize reactor power on a short time scale.

On the other hand, too large negative void coefficient could cause oscillatory instabilities and might lead to the need for rather high reactivity rates from the reactivity control system in order to achieve the highest rates of change of electrical output demanded in normal operation of the plant.

A useful negative void coefficient may be achieved by under-moderating the reactor.

PRESSURE COEFFICIENT OF REACTIVITY

The pressure coefficient of reactivity is defined as the change in reactivity per unit change in pressure:

- *The pressure coefficient of reactivity for the reactor is the result of the effect of pressure on the density of the moderator (sometimes called moderator density reactivity coefficient).*
 - As pressure increases, density correspondingly increases, which increases the moderator-to-fuel ratio in the core.
- *In a typical under-moderated core, the increase in the moderator-to-fuel ratio will result in a positive reactivity addition.*
- *In reactors that use water as a moderator, the absolute value of the pressure reactivity coefficient is not a major factor because it is very small compared to the moderator temperature coefficient of reactivity.*

EFFECT OF TEMPERATURE ON POWER TRANSIENTS

How do the temperature feedback mechanisms affect a reactor power transient?

Example: gas-cooled, graphite-moderated reactors

Assumptions and simplifications:

1. The reactor flux shape remains unchanged throughout the transient:

$$\nabla^2 \phi \propto \phi \text{ for all values of time and no spatial effects.}$$

2. No delayed neutron effects.

EFFECT OF TEMPERATURE ON POWER TRANSIENTS

Scenario:

- The reactor is running steadily at nominal power when the control rods are moved slightly to cause a small increase in multiplication dk .
- For a few seconds, the effects of the change in power on reactivity will be insignificant.
- Subsequently, the rising fuel temperature will begin to reduce the net reactivity (negative fuel temperature coefficient).
- Decrease of net reactivity will arrest the power rise.
- After some time (determined by the time constant of the fuel τ_f) the power will stabilize at a level, which is determined by what is reactivity drop due to increase in fuel temperature needed to compensate the initial reactivity excursion.
- At this stage, the equation $\delta k + \alpha_f \Delta T_f = 0$ is obeyed to a fair approximation.

ΔT_f is the change in statistically weighted mean fuel temperature

α_f is the fuel temperature coefficient of multiplication

EFFECT OF TEMPERATURE ON POWER TRANSIENTS

Additional simplifications:

3. Neglect the time constants of fuel and coolant.
4. Neglect changes in the temperature of the coolant gas entering the core.

It can be shown (see class notes) that:

$$\delta P = -\frac{\delta k}{(\alpha_f \Delta T_f^* + \alpha_m \Delta T_m^*)} \left\{ 1 + \frac{\alpha_m \Delta T_m^*}{\alpha_f \Delta T_f^*} e^{-\omega t} \right\}$$
$$\omega = -\frac{(\alpha_f \Delta T_f^* + \alpha_m \Delta T_m^*)}{\alpha_f \Delta T_f^* \tau_m}$$

➡ The condition for instability is clearly $\omega < 0$

EFFECT OF TEMPERATURE ON POWER TRANSIENTS

- Since $\alpha_f \Delta T_f^*$ is always negative in practice, the instability criterion may also be written

$$\alpha_f \Delta T_f^* + \alpha_m \Delta T_m^* > 0$$

- !! This is an important result that illustrates the fact that it is the weighted sum of the feedback (reactivity) coefficients, which determines whether or not the net feedback is positive.
- !! The weighting depends upon the strength of the coupling between temperature and power, which is reflected in the steady state material temperatures.
- !! The fuel temperature is more sensitive function of neutron flux or power since it changes more in a transient - it thus has a higher weighting.

TEMPERATURE COEFFICIENTS OF REACTIVITY

In summary:

- A negative temperature coefficient of reactivity is desirable because it makes the reactor more self-regulating:
 - *An increase in power, resulting in an increase in temperature, results in negative reactivity addition due to the temperature coefficient.*
 - *The negative reactivity addition due to the temperature increase will slow or stop the power increase.*
- The fuel temperature coefficient is more effective than the moderator temperature coefficient in terminating a rapid power rise:
 - *The fuel temperature immediately increases following a power increase.*
 - *The moderator temperature does not increase for several seconds.*
- Reactors are usually designed to operate in an under-moderated condition so that the moderator temperature coefficient of reactivity is negative.

TEMPERATURE COEFFICIENTS OF REACTIVITY

In summary (cont.):

- The pressure coefficient of reactivity is usually negligible in reactors moderated by subcooled liquids because the density of the liquid does not change significantly within the operating pressure range.
- The void coefficient of reactivity becomes significant in a reactor in which the moderator is at or near saturated conditions.
- *Since the fuel temperature coefficient is always negative, we can write the criterion for unstable reactor as:*

$$\alpha_f \Delta T_f + \alpha_m \Delta T_m > 0$$

MODERATOR TEMPERATURE EFFECT

TMI-1 Main Steam Line Break (MSLB)

- ➡ Break occurs in one steam line upstream of the cross-connect.
- ➡ Control rod with maximum worth is assumed stuck out.
- ➡ Event is characterized by significant space-time effects in the core due to the asymmetric cooling.
- ➡ Conservative assumptions utilized to maximize RCS cool-down.
- ➡ Major concern: possible return-to-power and criticality.

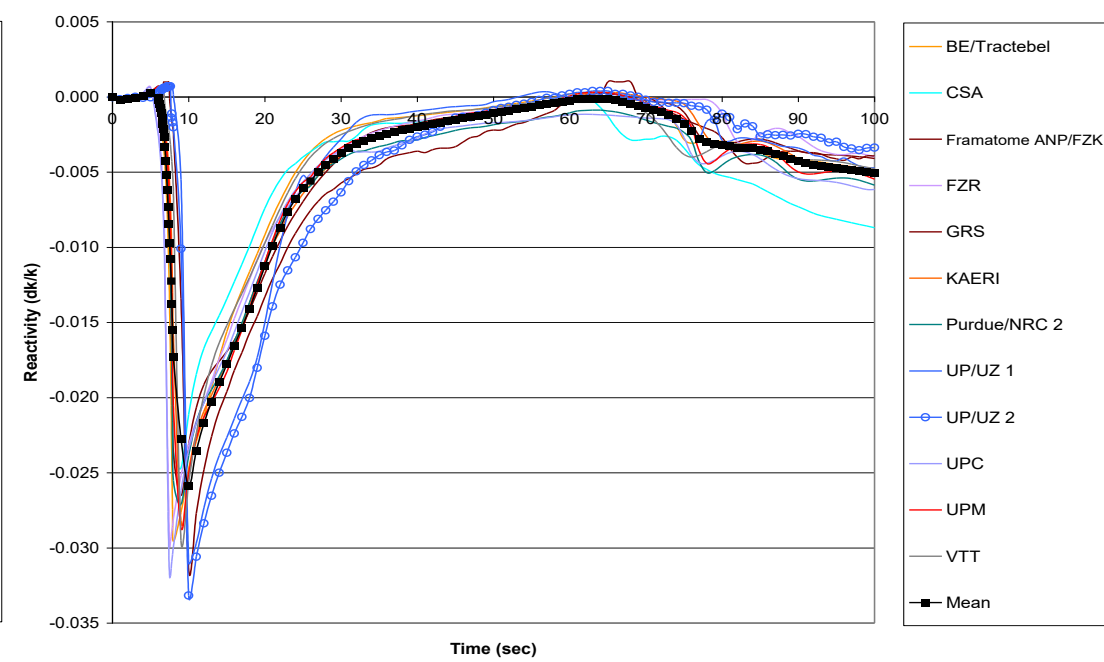
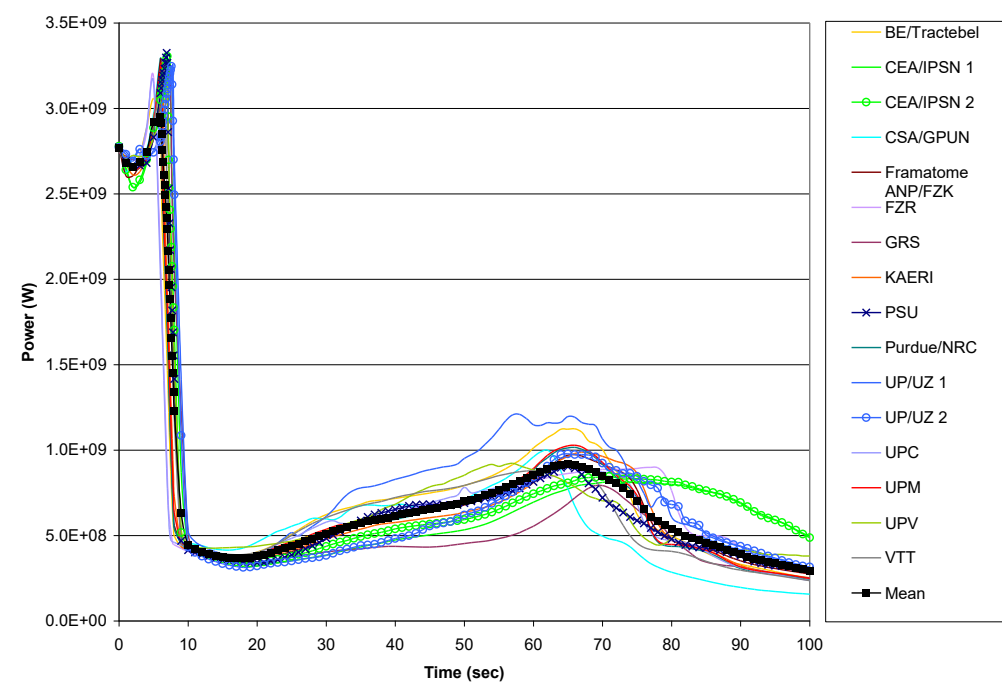
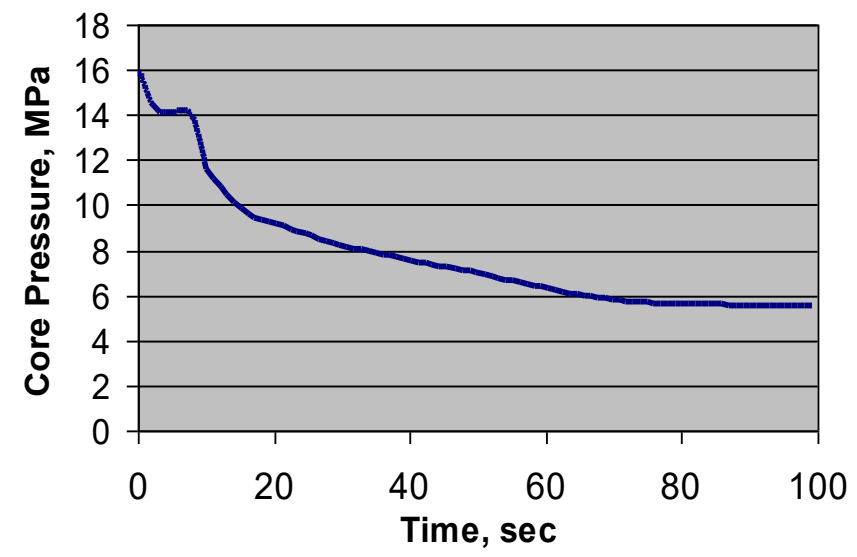
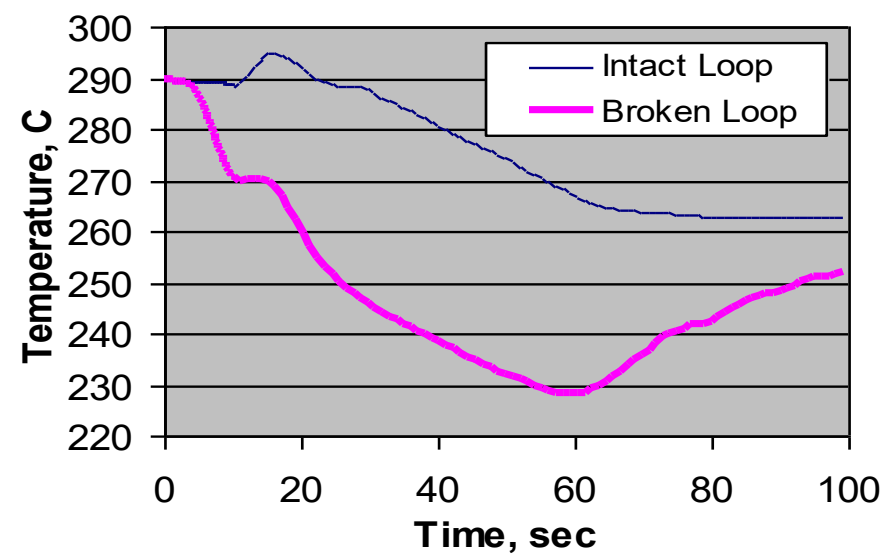
MODERATOR TEMPERATURE EFFECT

TMI-1 Main Steam Line Break (MSLB)

- ➡ Break occurs in one steam line upstream of the cross-connect.
- ➡ Control rod with maximum worth is assumed stuck out.
- ➡ Event is characterized by significant space-time effects in the core due to the asymmetric cooling.
- ➡ Conservative assumptions utilized to maximize RCS cool-down.
- ➡ Major concern: possible return-to-power and criticality.

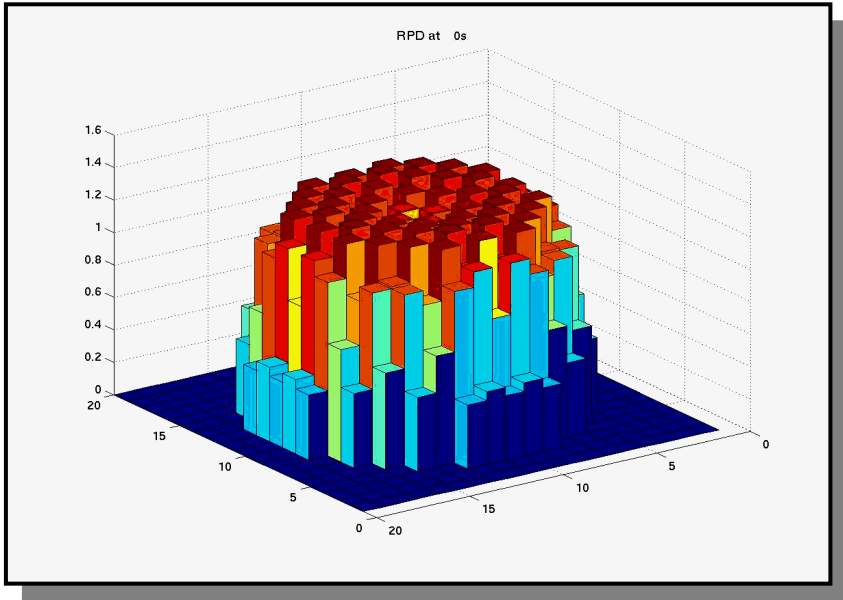
MODERATOR TEMPERATURE EFFECT

TMI-1 Main Steam Line Break (MSLB)



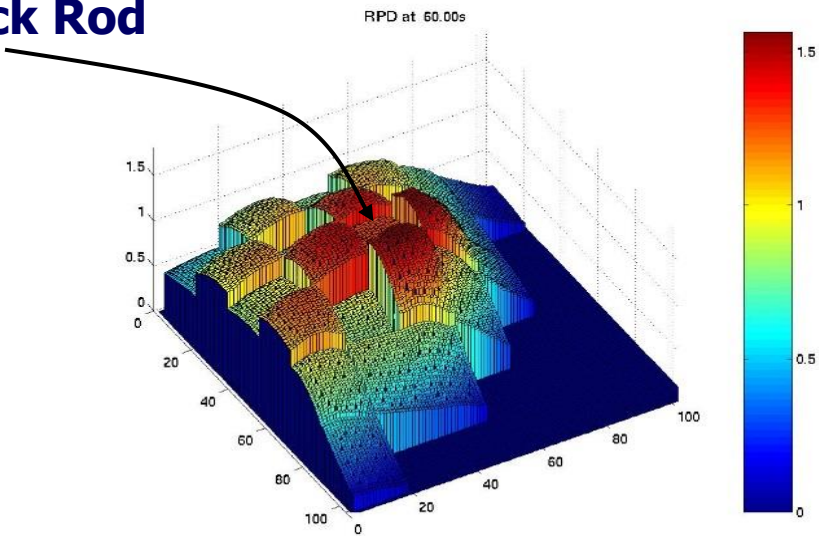
MODERATOR TEMPERATURE EFFECT

TMI-1 Main Steam Line Break (MSLB)



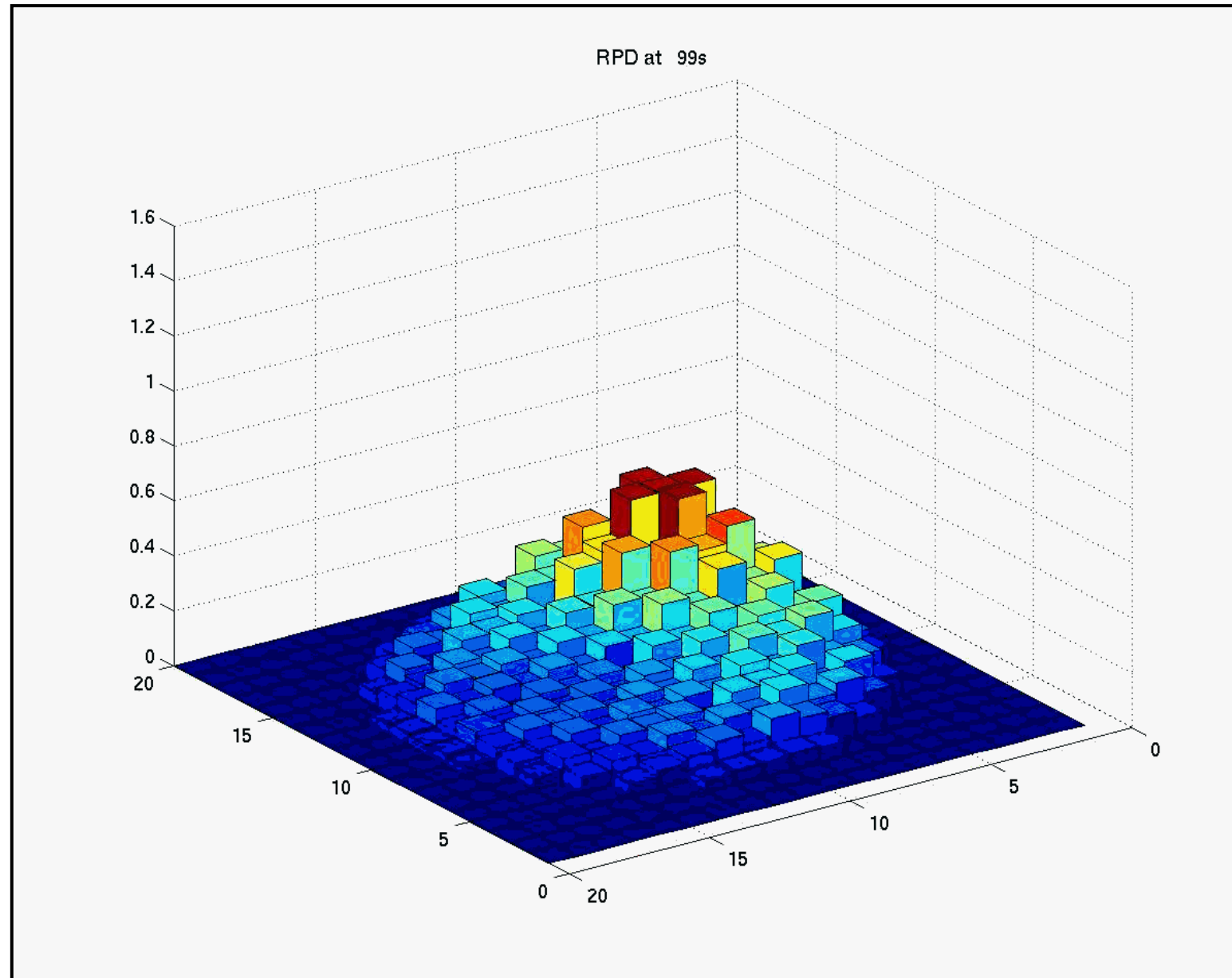
Pin Power in Vicinity of Stuck Rod

Stuck Rod



MODERATOR TEMPERATURE EFFECT

TMI-1 Main Steam Line Break (MSLB)



Borax - Safety experiment on a boiling water reactor

<http://www.youtube.com/watch?v=8WfNzJVxVz4>

Next Class

BWR Instabilities & General Reactor Stability